

# Tanvi Raut

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## EDUCATION

North Carolina State University, Raleigh, NC

Aug 2023-May 2025

Master of **Computer Science with GPA of 3.76/4**

Coursework: Algorithms, Software Engineering, Computer Networks, DBMS, Neural Networks, Big Data Analysis

Vishwakarma Institute of Information Technology, Pune, India

Aug 2019-May 2023

Bachelor of Engineering in **Information Technology with CGPA of 9.56/10**

## TECHNICAL SKILLS

**Languages:** Java, Python, SQL, C++, JavaScript, CSS, HTML, TypeScript

**Tools:** Slack, Git, Jira, Agile SDLC, CI/CD, RESTful APIs, Postman, Linux, JUnit, .Net, Apache JMeter, Selenium

**Frameworks:** Node.js, Bootstrap, ReactJS, Django, PyTorch, Apache Hadoop, Apache Spark, Automation Anywhere

**Databases:** MySQL, PostgreSQL, NoSQL, Snowflake, MongoDB, Teradata

**Analytical Tools:** PowerBI, Tableau, R, Visual Studio, Jupyter, Excel.

## WORK EXPERIENCE

Research Assistant, North Carolina State University, NC

Jan 2025 - Ongoing

- Identified and integrated open source LLMs into existing course chatbot, enhancing contextual **accuracy by 67%**.
- Increased response accuracy **by 40% by implementing Retrieval-Augmented Generation (RAG)** and vector databases for better retrieval.

Software Engineer Intern, NetApp – Research Triangle Park, NC (Hybrid)

Jun 2024 - Aug 2024

- Forecasted Annual Recurring Revenue (ARR) with **92% model accuracy using time-series** forecasting techniques in Python and SQL.
- Automated reporting via Power BI and Excel macros, **saving 4+ hours/week** and reducing manual work by 20%.
- Optimized data ingestion using Excel functions and **Power BI**, cutting refresh time by 30% and **enabling real-time KPIs**.
- Upgraded legacy BI systems with **ML-driven analytics**, improving forecast accuracy and visibility by 25%.

Software Engineer Intern, Radiance Technologies LLC (Remote)

Aug 2022 - Jan 2023

- Developed web based cloud solutions using **JavaScript, ReactJS, and MongoDB**, as per customer requirements.
- Executed comprehensive **tests on REST APIs** using POSTMAN for data fetching across 10+ deployed websites.
- Collaborated with internal stakeholders to create automation statistics visualization and reports using **Power BI** with **65+ feature which improved KPIs by 16%**.
- Implemented Robotic Process Automation bots for testing, improving web application **testing efficiency by 60%**.

Research Intern, VIIT - Maharashtra, India (On-site) [link](#)

Aug 2022 - Dec 2023

- Integrated **MobileNetV2 with LSTM** model for real-time CCTV violence detection, triggering alert to authorities.
- Curated violence video dataset from open-source platforms achieving **94% accuracy**, surpassing previous models.

## PUBLICATIONS AND PROJECTS

Plant Seedling and weed management - Agriculture initiative (EfficientNet, ResNet, MobileNet, XGBoost)

- Developed a model leveraging **Vision Transformers (ViT) and EfficientNet** for plant seedling classification.
- Enhanced accuracy with segmentation, bilateral blurring, LAB color space conversion to improve dataset quality.
- Achieved 95.32% test accuracy, surpassing classifiers like ResNet, MobileNet, and XGBoost in precision.

Autonomous Lane Detection (Computer Vision | YOLO) [link](#)

- Developed a lane detection model using **YOLO, Canny edge detection, and Hough Transform** for real-time autonomous navigation. Optimized **image processing pipelines** and tested on diverse datasets.

Intent Analysis - Market Intelligence : Facile Info-Serv Private Limited (SPONSORED PROJECT)

- Developed an intent determination model **with 92% accuracy using BERT, RoBERTa, and Scikit-learn for B2B**.
- Built full-stack application using React, Node.js, Express.js, integrated ML model for real-time intent detection.
- Designed RESTful APIs to process text data and return predictions, improving system **response time by 30%**.

Covid 19 Data Analysis

- Analyzed and manipulated India's Covid 19 data for the year 2021 to enhance vaccination process.
- Performed data pre-processing and created data visualizations for provincial statistics on active cases, casualties and doses per state using **Python and Tableau**.
- Determined **9 unique recommendations** that could help the Federal Government reduce the cases by 20%.

## ORGANIZATION

- EDC organization:** Led front-end development, content strategy, and business decisions for multiple projects over tenure of two years.